



Play, watch, analyze, repeat: How do players develop competitive gaming skills?

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ABSTRACT

Players usually have a strong desire to improve their game-specific skills to perform well in a game. A desire which has been further reinforced by the increase of multiplayer games and the rise of competitive gaming. While research on learning through games is extensive, work on how players gain expertise in competitive games is less common. This paper presents an exploratory mixed-methods study examining how players learn to play video games across three competitive game genres and which factors influence their self-efficacy. Our results show that the approaches to learning as well as influential factors on self-efficacy are largely independent across the investigated genres. Moreover, learning to play extends beyond the act of play itself and is strongly reflective and social. Based on the results, we discuss high-level characteristics of learning and self-efficacy which can serve as a starting point for future research and design interventions aimed at supporting learning processes of players.

1. Introduction

With steady advances in technology, modern video games have become incredibly complex systems containing large amounts of information and interacting variables. While people find enjoyment in navigating these highly interactive worlds with seemingly endless possibilities, making sense of these complex environments and systems with often overwhelming amounts of information can be challenging and may require extended commitment. Over the years – strongly fueled by the rise of competitive gaming, esports, and publicly available access to in-game data – players have developed an interest in ways to improve their game-specific skills. This increased interest is evident in the number of available websites dedicated to performance analytics, replay tools, streaming platforms, and similar (see also [1,2]). While data-driven tools are abundant, Kleinman and Seif El-Nasr [2], however, identified a lack of empirical work that studies how players actually use their data to gain expertise in esports. This not only holds true for esports specifically but also for learning in games in general. As Hesketh et al. [1] discussed, research on learning and games majorly focused on how games can facilitate learning but less on how players

learn to play a game. While research on the latter is gaining ground, our current knowledge on how players acquire gaming skills and construct knowledge about game mechanics is still limited. These are, however, vital questions as many have argued that games are all about learning and keeping the player learning (e.g., [3,4]).

Learning to play can be guided by a game (e.g., via tutorials) but can also be largely self-regulated with players setting their own performance goals, instructing themselves, or continuously self-monitoring their performance. However, self-regulated learning is influenced by various factors with self-efficacy being regarded an important one (cf. [5]). Following Bandura [6], self-efficacy can be summarized as the personal judgment of how well one can execute courses of action required to deal with prospective situations. Since the concept was proposed, it found widespread adaption in psychology, health, sociological research, or also in political science literature [7]. Self-efficacy beliefs are central to Bandura's own social cognitive theory [8] – which posits, briefly summarized, that an individual's knowledge acquisition takes place in a social context through interactions with others – but also

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found application, for instance, in sociological research around socioeconomic status. Other disciplines have postulated specific self-efficacy types such as the political sciences which introduced the concept of political efficacy [9] to refer to an individual's beliefs to influence political processes. In the context of games, self-efficacy can be viewed as *the judgment of a player about his/her capabilities to organize and execute the courses of action required to attain designated types of performance* following Kordyaka et al.'s [10] slight adaption of Bandura's definition. Self-efficacy influences the goals individuals set, their commitment and effort to achieve these goals, and their perseverance and resilience to adverse situations [8]. Hence, self-efficacy is critical for human learning as it can directly affect motivation [11]. In the context of gaming, it can further increase the likelihood to engage with a game for longer time periods and persist in situations of failure and adversity [12], which we deem to be especially relevant for learning to play competitive games. While playing games, players with high self-efficacy will believe in overcoming adverse challenges and put effort into mastering the game, while players with low self-efficacy may doubt their abilities and skills, exerting lower effort or withdraw when faced with adverse challenges. However, self-efficacy in relation to games has mainly been explored from the perspective of how games can foster self-efficacy outside of them (e.g., [13–15]).

To help fill the above mentioned gaps we have conducted a mixed-methods study, pairing qualitative interviews and quantitative surveys, to explore how players learn to play games and which factors influence self-efficacy beliefs. While others have looked into how specific aspects, for instance, watching and spectating others (e.g., [16,17]) or how visualization tools (e.g., [18–20]) can contribute to understanding game mechanics, our study is focused on learning to play in general in the vein of the preliminary study of Hesketh et al. [1] and the work of Iacovides et al. [21]. Recent research studies have shown that learning, learning strategies, and cognitive functioning while playing video games is affected by game genres [22–24]. It thus seems reasonable to assume that the approaches to learn are also influenced by overall genre characteristics (e.g., team-based vs. one-against all, features that afford learning, etc.). Hence, to observe potential differences with respect to learning and self-efficacy across genres, we investigate these constructs through the lens of three popular competitive genres (multiplayer online battle arena, battle royale, and real-time strategy). Our study is based on a corpus of 16 interviews of players of three competitive online games one for each genre: *Defense of the Ancients (DotA 2)* [25], *PUBG: Battlegrounds* (previously known as *PlayerUnknown's Battlegrounds*,) [26], and *StarCraft II* [27]. The developed categories were then quantified through an online survey with a mainly male sample ($n = 264$) expanding the scope of considered games within the above genres. In summary, our study explores the following two research questions and contributes to our understanding of:

RQ1: How do players learn to play across the three competitive genres of RTS, MOBA, and BR.

RQ2: Which aspects can impact self-efficacy (positively or negatively) across the different genres?

Based on the results, we discuss high-level characteristics of how learning and self-efficacy manifest themselves in these genres. Together with the developed categories, these can serve as useful guideposts for implementing teaching practices, systems, and tools for learning to play in- and outside of competitive games. In summary, our study contributes to our understanding of how players learn to play by expanding findings of previous works but also reaffirming certain previous results. Iacovides et al. [21] as well as Pelletier and Oliver [28] both relied on observational analysis of (videos of) gameplay, thus focusing on how learning happens through play. However, games have become social ecosystems (including community websites, forums, etc.) and it is equally interesting to explore how these are used by players to improve their gaming skills. Thus, we take a broader perspective and also look

into resources players refer to outside of the game. As such we employ a combination of interviews and surveys to capture a wider variety of factors which would not be possible with an approach purely relying on observing play itself. In a sense, this is similar to Hesketh et al. [1] aims but they focused on high-level analytical aspects (processes, tools, and goals). In contrast, our study provides a more comprehensive investigation on the specific strategies, tools, and resources players are using. Additionally, we not only rely on a qualitative inquiry alone to identify factors as the works above but also quantify their relevance with a large sample of players.

2. Related work

Work on game-based learning has shown that games have the potential to yield positive learning outcomes across a range of subjects (see, e.g., [29,30]). Others, such as the seminal work of Gee [31] or recent work of Barr [32,33], have documented how commercial video games initially created for entertainment, can facilitate learning and contribute to the development of real-world competencies. In contrast, our focus of interest is to explore which strategies and resources players use to develop their in-game skills. Thus, to keep the discussion focused, we specifically review literature addressing aspects of how people learn to play games (RQ1). Secondly, we discuss previous work considering the role of self-efficacy in games (RQ2).

2.1. Learning to play

Kleinman and Seif El-Nasr [2] reflected on the need for more player-centric research to understand how players (a) use data to gain expertise in esports and (b) what data they need. Towards this end, the paper at hand is specifically focused on the former of these two aspects. Probably the most related works to ours in this regard are the ones by Hesketh et al. [1] and Iacovides et al. [21]. Hesketh et al. [1] reported on preliminary findings from a grounded theory study to inquire into how amateur and semi-professional players learn to play team-vs-team games, specifically *DotA 2* and *Counter Strike: Global Offensive*. Their analysis revealed three analytical aspects: learning processes, learning tools, and learning goals. Our study is similar in terms of goals but includes other games (apart from *DotA 2*) and also attempts to quantify the importance of different aspects across different genres. Similarly, using observational studies of players, Iacovides et al. [21] identified a number of strategies to achieve *breakthroughs* (i.e. instances in which learning occurs) in games. Their results show that these strategies also vary based on if a game is single player or cooperative. As we will discuss in the course of this paper, our results partly align with the above two categorizations, confirming previous findings and suggesting a certain transferability and generalizability of the results. Pelletier and Oliver [28] proposed a methodology to analyze instances of play through observation under the lens of activity theory to understand how a person plays a game. Nagorsky and Wiemeyer [34] have developed a framework of relevant competencies and training areas for esports, showing that competencies are game-specific.

Focusing specifically on young children (7 to 10 years), Blumberg and Sokol [35] investigated which strategies they use to learn to play an unknown video game, distinguishing between internal and external (i.e. referring to sources external to the individual such as asking for help or observing someone else play) strategies. The identified strategies were trial-and-error, watching others, and read or ask for instructions. Older children and more frequent players more likely relied on internal strategies. Iacovides et al. [36], concerned with players more generally, made a similar distinction into learning through play itself and external resources and interactions. Focusing on a similar age group (fourth- and fifth-grade children) as Blumberg and Sokol [35], Hamlen [24] identified six learning strategies children apply (e.g., repetition, exploration, using cheats, observation) and found that their

importance partly depends on the genre. Repetition was rather applied for action games while imagination for adventure games.

Others have taken a less general approach and rather looked into specific approaches to learning to play. For instance, spectating games has also been repeatedly connected with developing game-related competencies. Georgen [37] argued that the act of watching live streaming provides opportunities for learning through *cognitive apprenticeship*, as watching games helps novices to, amongst others, observe expert strategies. Live streams, following Georgen further, highlight game mechanics, technical skills, and expert strategies which otherwise would require considerable practical experience to develop. Indeed, a study by Hamilton et al. [38] has shown that the desire to learn about games is a common reason to start watching streams. However, a more recent study by Sjöblom and Hamari [16] paints a more nuanced relation between the need to learn about games and watching streams. Cognitive motivations such as information seeking showed only a small positive association, with the authors arguing that the live aspect of streams may be detrimental to learning. In another study employing the *Sports Consumption Motivation Scale* [39], Hamari and Sjöblom [17] found that the motivation to acquire knowledge to be a positive determinant for spectating esports. These studies were aimed to understand the reasons for spectating games while we are focused on exploring the different ways players use to develop game-related skills.

Others have specifically focused on how visualizations can help facilitate training and skill development. Hazzard [3] considered training visualizations as one of the two primary purposes visualizations serve for players (the other being visual status representations). Bowman et al. [40], proposing a design framework of visualizations for games, provided a more fine-grained classification of the purposes fulfilled by visualizations, including training. Medler [41] emphasized the usefulness of so-called *player dossiers* (that is, data-driven reporting tools that use player's gameplay data) to learn to play from past experiences. These papers discuss the usefulness of visualizations for training from a theoretical rather than empirical perspective. Their findings can be used to inform the design of visualizations to ensure they are useful and appropriate for players. They are, however, focused on the 'what' rather than the 'how'.

2.2. Self-efficacy and games

Self-efficacy refers to one's belief to effectively perform a given task (cf. [6]) and has been shown to be important for learning [42] and motivation [11]. As such self-efficacy has particularly been examined in teaching environments such as teacher self-efficacy [43], classroom learning [44], and higher education [45,46]. It has, however, also been studied in other contexts where performance or motivation are vital such as in workplace environments [47], health behavior [48,49], and – probably most closely related to our context – sports. A meta-analysis of 45 studies related to sports [50] showed an overall positive and moderate relationship of self-efficacy beliefs and performance, however, the relationship was not homogeneous across the investigated studies. A more recent meta-analysis by Lochbaum et al. [51] corroborated these findings, also showing an overall positive and moderate relationship, but also highlighting that high self-efficacy might not always improve sports performance. Given these potential positive effects in sports, understanding which factors impede or increase performance in competitive games can be valuable to better design for skill development.

Besides, literature (e.g., [11]) has found self-efficacy to be one of the strongest predictors of motivation. Klimmt and Hartmann [12] argue that this also holds true for continued activity with video games as it is reasonable to assume that players constantly reevaluate their chances of success. Players with stronger game-related self-efficacy beliefs, according to Klimmt and Hartmann [12], are thus likely to engage with games for longer periods of time. As such investigations into factors that can improve or be detrimental to self-efficacy with

respect to game playing, as the one reported here, can offer valuable insights. Vorderer et al. [52] investigated if higher game-related self-efficacy leads to improved likelihood of selecting competitive games. Results, although having weak effects, indicate that the motivation to compete as well as self-efficacy are both predictors of the general time of play. The motivation to compete has, however, been shown to be a stronger predictor for playing competitive games than for a player's self-efficacy. Our study aims to understand players' view on self-efficacy in the context of learning to play competitive games.

In general, Bandura [11] describes four sources for developing self-efficacy: mastery experiences, vicarious experiences, verbal persuasion, and physiological and emotional states. It can be assumed that games that promote these sources facilitate building self-efficacy which, in turn, helps to create a more satisfied and motivated player base, contributing to the longevity of a game. This requires a better grasp of factors that influence how gaming-related self-efficacy is formed. However, to the best of our knowledge, these factors have not been studied yet in depth. Our work thus contributes to filling this gap by presenting a first step into this direction.

Additionally, following Bandura's social cognitive theory [8], self-efficacy could be further positioned as an internal disposition which does not operate in isolation, but in a reciprocal interaction with other behavioral and environmental determinants in the societal structures. Games are increasingly social constructs where players interact with each other in-game but also outside. Consequently, the role of community interactions becomes interesting in shaping self-efficacy. Through our study, we aim to shed more light on how self-efficacy is shaped in the context of multiplayer competitive games which have active online communities.

It is also worth noting that work on self-efficacy in relation to games has primarily focused on how games can contribute to out-of-game self-efficacy. For instance, Santos et al. [13] and Song et al. [15] saw evidence that exergames promote exercise self-efficacy. Our focus is different, as we are interested in unearthing potential factors that impede self-efficacy with respect to gameplay itself.

Lastly, it should be acknowledged that self-efficacy and learning have been found to be strongly interrelated (see, e.g., [5,53]). For instance, Hayat et al. [54] reported that medical students' self-efficacy had an impact on their metacognitive learning strategies while Magogwe and Oliver [55] observed a significant (albeit weak) relationship between self-efficacy beliefs and the use of language learning strategies. Given the documented cross-domain evidence, it appears likely that a similar interrelationship could surface in relation to the learning of games. Understanding of self-efficacy influencing factors and the different approaches to learning to play in the first place can be a useful initial step in this direction. Thus, while exploring such an interrelationship is beyond the scope of this paper, our results can contribute to future research efforts in this direction.

3. Study procedure

The findings discussed in this paper are part of a two-part study exploring (1) information needs with respect to post-play data visualizations and (2) learning and skill development in games. Both parts are based on the same underlying dataset but focus on different parts of the conducted interviews and survey and have been analyzed with respect to different research foci. This paper focuses on part (2) while results of part (1) have been published previously.

The study was approved by the ethical review board of the university and followed a mixed-methods approach consisting of two steps:

1. interviews with players with varying levels of playing expertise to develop relevant categories with respect to RQ1 (learning) and RQ2 (self-efficacy) which were subsequently
2. quantified using an online survey.

This approach was chosen in view of our goal to capture both in- and out-of-game factors related to RQ1 and RQ2. The aspects were explored with respect to three popular competitive gaming genres, namely multiplayer online battle arena (MOBA), battle royale (BR), and real-time strategy (RTS). We focused on these as they are inherently skill-based and thus developing these skills takes on an important role.

To keep the number of interviews from becoming too large while still ensuring a reasonable number and sufficient range of expertise, we decided to recruit players from one representative game of each genre (i.e. *DotA 2*, *PUBG*, *StarCraft II*) known to have an active player community. For the survey we then expanded the range of included games to quantify the developed categories on a genre-basis. This allowed us to explore our research questions in depth and with breadth.

3.1. Games and genres

In the following we describe the three investigated genres along with the games focused on during the interviews. It is worth noting that genre classifications for video games have their limitations since games within the same genre can still vary based on individual characteristics [56]. However, as we are rather interested in a high-level conception of learning and self-efficacy we – based on the suggestion by Clarke et al. [56] – use Wittgenstein's idea of family resemblance to conceptualize genres as the presence of a critical, identifiable mass of shared characteristics, instead of all attributes needing to be present.

MOBA: MOBAs such as *DotA 2* [25] are strongly team-based and require players to cooperate closely with their teammates together with effective team composition to achieve victory. MOBAs feature numerous playable characters (often referred to as heroes or champions), all with their unique strengths and weaknesses which can also be further customized in several ways. Players have to be aware of these manifold characteristics and efficient hero builds to play efficiently.

BR: Battle royale games such as *PUBG* [26] pit a player against all others until the last player remains alive. Players have to roam the game's map for better gear, loot, and weapons to increase chances in combat. To ensure that the action remains tense over the course of a match, the playable area constantly shrinks over time.

RTS: In real-time strategy games such as *StarCraft II* [27], players control numerous units to compete against each other for certain objectives. RTS games usually require players to gather resources which are necessary to improve infrastructure and buy new units. Players typically have to choose among different factions which usually have specific strengths and weaknesses. This part is particularly strong in *StarCraft II* where the contending factions require a different style of play. *StarCraft II* also puts considerably focus on establishing a strong economy which has resulted in players compiling so-called 'build orders' in which different buildings should be constructed.

3.2. Interviews

Design. Participants for the interviews were selected based on a screener distributed on *subreddits* and *Discord* servers related to our three investigated games to cover diverse gaming experiences and views. Interviews were conducted online in April and early May 2020, lasted for about one hour, and were semi-structured. Before the interview, the topic was introduced and consent for audio recording was obtained. Predetermined questions of relevance with respect to the purpose of this paper included, for example: *How do you learn and improve your skills in the game?*, *Which graphical elements of the interface do you use for decision making?*, *How helpful are they to improve your skills?*, *Which information or feedback can increase self-efficacy?* Two trial interviews were held to ensure clarity of the interview questions.

Participants. Out of 30 volunteers, 16 were invited for interviews. The selection of the 16 interviews was made on a convenience basis, taking into account the breadth of expertise, all three types of games, and the availability of the interviewees within the given time. Seven participants were interviewed for *DotA 2*, six for *StarCraft II*, and three for *PUBG*. In terms of demographics, 15 were male while one preferred not to disclose the gender. Age ranged from 20 to 29 years (mean = 23.75, SD = 2.8) and all had multiple years of gaming experience. When asked to self-describe them, two participants self-identified as ex-professional players, two as former semi-professional players, and four as being active community members. Six mentioned to play games more for the social aspect, and three considered themselves as highly competitive players. All the interviewees seemed aware of the concept of self-efficacy. However, to ensure that there was a common understanding, it was introduced during the interview – as a simplified form of Bandura's definition [6] (cf. Section 1) – as *how a person views their own ability to successfully perform a specific task*.

Analysis. Recordings of the interviews were transcribed and analyzed using an inductive qualitative content analysis process to derive categories in respect to our research questions. For that purpose, all transcripts were coded to develop a first set of categories. Through discussion with a second researcher the categories were refined twice. The interviews were then re-coded using the final set of categories. Afterwards, a third researcher coded four (out of a total of 16) interviews – selected randomly but ensuring that each genre is represented once – independently based on the final categories. Four interviews were chosen based on literature [57] suggesting that multiple coding of a proportion of 10% to 25% provides a trustworthy estimate of inter-coder reliability (ICR). Calculation of ICR using percentage agreement and kappa indicates good to excellent agreement for all categories (κ usually > 0.71 except for a single category – COPYING OTHERS, one of the least mentioned aspects in the interviews – with $\kappa = 0.59$; percentage agreement ranging from 97.9% to 100%), showing good internal consistency of the categories. Resulting counts presented in Tables 1 and 2 are based on sentences that discuss a given topic. Subsequent sentences discussing the same topic were counted as one instance.

3.3. Survey

Design. The survey was conducted over a period of 40 days from late May to July 2020 via *Google Forms*. After obtaining consent, the survey collected basic demographics (age, gender), the name of the game for which a participant was taking the survey, and information on gaming experience (years playing the game for which the survey was filled, hours spent playing the game in the 30 days prior to taking the survey). Afterwards, participants were asked to indicate how much they agree with statements reflecting the developed categories with respect to learning and self-efficacy (e.g., *Copying what other people do is a good way to learn*; *Playing with friends is a good way to learn*; *Showing people their impact on a game can increase self-efficacy*; *Showing people how fair a match-up was can increase self-efficacy*) on a 7-point Likert-like scale anchored by 1 = *totally disagree* and 7 = *totally agree*. The survey was distributed among *subreddits* and *Discord* servers related to the three covered genres.

Participants. The survey received 292 responses of which 264 remained for analysis after removing careless responders based on longstring analysis [58] and excluding responses related to games not within the three explored genres. Gender and age were distributed as follows: 249 were male, 10 female, 1 non-binary, and 4 did not report their gender; 77 were below 20 years, 153 where between 20 and 29 years of age, and the remaining 34 were older than 29 years (mean = 23.0, SD = 5.8). 83 participants (31.4%) answered the survey for BR games (*Fortnite* = 60, *PUBG* = 23), 80 (30.3%) with respect to MOBAs (*League of Legends* = 58, *DotA 2* = 21, *Heroes of the Storm* = 1), and 101

Table 1

Categories regarding relevant aspects for learning and skill-development in a game (#P = number of participants mentioning it, #S = number of statements).

Category	Description	#P	#S
REPLAYS	Mentions of watching their own replays as a way to learn, gain insights, and identify mistakes.	14	38
FAMILIARIZATION	Statements about the importance of playing and actually experiencing how the game works to build intuition and a sense of the game.	14	31
CONSCIOUS REFLECTION	Mentions of conscious trial-and-error in specific situations by purposefully changing one variable in a situation and assessing the outcome in order to find out what works and what does not. This relates to Korthagen's model of reflection, a circular process consisting of five steps: 1. act and experience, 2. reflect, 3. identifying essential aspects, 4. develop alternatives and choose, and 5. repeat.	9	18
FRIENDS	Statements about playing with friends and learning from them.	9	14
WATCHING GAMEPLAY	Mentions of watching gameplay videos from either pro tournaments or casual games on <i>Youtube</i> , etc. to observe or study diverse gameplay aspects	8	14
BUILDS AND TACTICS	Statements on finding specific builds or tactics online and following these. This includes sequencing, choice of buildings, items, leveling/upgrading, or similar.	6	15
ROUTINE AND HABITS	Statements emphasizing the value of creating routines and habits, both mental and physical, for various reasons. These include: taking load off the player's working memory, optimizing efficiency, and developing an attitude that persists, etc.	5	13
COPYING OTHERS	Mentions of copying other players' behavior to learn through replication of actions.	4	6
PHYSICAL DEXTERITY	Statements highlighting the importance of purposefully training certain skills.	3	9
EXTERNAL DATA TOOLS	Mentions of consulting data tools outside of what the game itself provides, e.g., statistics websites such as <i>OpenDota</i> [59] or <i>Sc2ReplayStats</i> [60].	3	6
CAMPAIGN	Mentions of playing through the campaign as a means of learning.	2	3
AI	Mentions of playing against the AI as a way to learn.	2	2

(38.3%) for RTS games (*StarCraft* series = 67, *Age of Empires* series = 31, *Command & Conquer* series = 3). Respondents indicated to have been playing the game for which they filled in the survey on average since 5.5 years (SD = 4.3, min = <1 year, max = 20 years).

Analysis. Ratings were treated as ordinal and analyzed using non-parametric tests. Differences between genres were examined with Kruskal–Wallis tests using Benjamini–Hochberg adjusted *p*-values to control for false discovery rate. In case of significance, post-hoc tests using Dunn's procedure were performed with a Bonferroni corrected alpha level of .05/3. In addition, we assessed if the years playing the game influenced the rating of the developed categories using logit-link ordinal regressions. Since the genres and their games have been on the market for different lengths of time and thus the range of years playing them varied between them (MOBA: 1 to 16 years; BR: <1 to 4 years; RTS: <1 to 20 years), we conducted this analysis individually for each genre.

4. Results

In the following we describe our results organized along our two research questions, namely how players learn to play (RQ1) and which aspects influence players' self-efficacy (RQ2). For each category, we first discuss the interview results followed by the survey responses. For easier identification, categories derived from the inductive analysis of the interviews (summarized in [Tables 1](#) and [2](#)) are written in slanted small caps. In addition, median scores of the relevance ratings of each category in the survey will be designated, for the sake of presentation, as: low (< 4), high (≥ 6), and as moderate otherwise. For detailed statistics, please refer to [Figs. 1](#) and [3](#).

4.1. How do players learn to play a game (RQ1)

Players identified a variety of mechanisms and resources that play a role in their learning process. As a fundamental mechanism, almost all (14 out of 16) interviewees pointed to the importance of building practice in the form of FAMILIARIZATION with a game. Players discussed the value of actually playing the game to gain experience as well as to develop intuition and a sense of the game. The following comment illustrates this well:

Experience will teach you repeating situations, that you can internalize. Because you had to make similar decisions before, you can make decisions later. [P2]

Being among the most frequently mentioned aspects in the interviews, the survey results also showed this to be highly relevant across all three genres (see [Fig. 1](#)) with no significant differences between them.

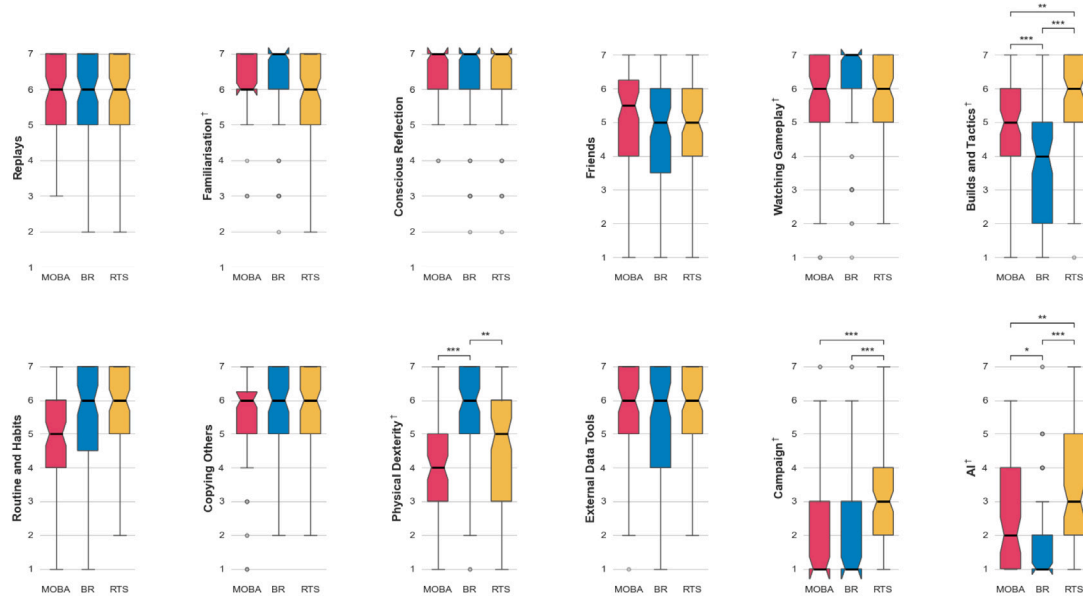
Players also mentioned that experience helped them to optimize hardware and software settings to suit their performance and to train their PHYSICAL DEXTERITY. This particular aspect was mostly highlighted by BR players in the interviews, which was also corroborated by the survey results where PHYSICAL DEXTERITY has been rated significantly more relevant for BR games, while having moderate relevance in MOBA and RTS games (cf. [Table 1](#)).

When discussing the importance of playing to learn a game, some players described how playing in different settings (i.e. FRIENDS, CAMPAIGN, AI) was useful for learning the game. Playing with FRIENDS was the most frequently mentioned mode and rated to be moderately relevant by players across the three genres with no significant differences between them (cf. [Table 1](#)). Interviews showed this to be especially useful

Table 2

Categories of factors considered by players to influence their self-efficacy (#P = number of participants mentioning it, #S = number of statements).

Category	Description	#P	#S
FOCUS	Statements on the importance of focusing on single aspects of the game and tracking progress according to these. This includes actively internalizing the prospected skill, freeing up working memory, as well as to create positive feedback and make the players feel good about their abilities.	9	24
TOXICITY	Mentions of toxic behavior in the scene, how this scares beginner players away, and how this could be handled or negated.	9	21
BEING OVERWHELMING	Mentions of the game being overwhelming in the beginning due to the sheer amount of information to be processed at the same time.	7	10
IMPACT	Statements on how feeling that one's actions and decisions matter and have impact on the game influences self-efficacy.	6	14
LEARNING CURVE	Mentions that the game has a steep learning curve and the beginning is just hard, and how realizing and accepting this is important for motivation and self-efficacy.	6	11
EXPLAINING STRATEGIES	Statements about how explaining concepts beyond the mechanics and controls can provide insights and directions for learning as well as help in identifying options and possibilities, in turn, creating more motivation and a sense of purpose.	4	7
DUNNING-KRUGER EFFECT	Statements related to the Dunning-Kruger effect [61], i.e. a cognitive bias of people with low ability at a task to overestimate their abilities.	3	4
MATCH FAIRNESS	Suggestions on quantifying and visualizing the fairness of the match-up to create reference for winning and losing and help people be okay with losing.	2	5



† Kruskal-Wallis H test significant at false discovery rate adjusted $p < .05$, Dunn's post-hoc test for multiple comparisons significant at $\alpha = .016$ (Bonferroni corrected), $**\alpha = .01$, $***\alpha = .001$

Fig. 1. Agreement on how relevant the developed categories regarding learning are across the different genres (1 = totally disagree, 7 = totally agree).

for beginner players who needed support and guidance from their friends (the challenges for beginner players due to the steep learning curve of these games are discussed in Section 4.2). The following comments emphasize the importance of playing with friends:

I would say you need some friends, people you play with often, to learn with and from. I don't think this can really come from yourself. [D6]

What personally really worked was playing with a set team, and having people tell you what you can do better, teaching you. [D3]

In addition to playing with friends, there were few mentions of playing solo CAMPAIGN and playing against the AI as a way of learning

the game. The survey results (cf. Table 1) also showed low importance for both of these modes to learn to play games belonging to the three genres. However, there were significant differences across the genres with both CAMPAIGN and AI being rated significantly more important for learning RTS games than MOBA and BR games. In addition, AI was rated significantly higher for MOBA than BR games, with AI being least important in BR games across all genres.

Other learning mechanisms focused on activities that players indulged into, before or after they played. This included watching REPLAYS of their own game to gain insights and identify mistakes. Players used different strategies while watching their REPLAYS to scaffold their learning such as switching between player's (own) perspective, enemy perspective, and spectator's view; isolating and focusing on one aspect

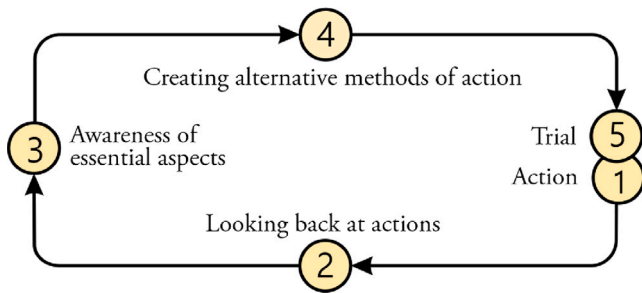


Fig. 2. Korthagen's ALACT Model of Reflection.
Source: Image adapted from [62].

of gameplay; and including teammates or others in analyzing the replays. The importance of watching replays is nicely encapsulated in the following comment:

However, watching a replay allows careful analysis of why and how. Ideally, you do this with other people, since their point of view is usually unbiased as opposed to your own, since you played the game and have more subjective information to cloud your judgment. [D3]

As the most frequently occurring category in the interviews, REPLAYS also had a high relevance for all the three game genres in the survey.

Furthermore, players showed a critical and reflective attitude, adopting a methodical approach which can be considered as CONSCIOUS REFLECTION [62]. That is, players learning through conscious trial-and-error by purposefully changing a specific variable in a situation and assessing the outcome to find out what works and what does not. This process resembles Korthagen's ALACT Model of Reflection (illustrated in Fig. 2). The following comment exemplifies this process well:

Sometimes, you find a major flaw in the macro strategy; then I usually start again with an adapted plan. Figure out what went wrong, change that, and then see what happens again. [S3]

CONSCIOUS REFLECTION was rated as having the highest relevance for learning, across all three game genres.

Interviewees also mentioned that WATCHING GAMEPLAY of other players is an important way to learn. While players watched videos from tournaments to casual games on streaming platforms such as YouTube, a number of interviewees stressed the value of watching professionals and experts as a source of inspiration. This was often accompanied by COPYING OTHERS and trying to replicate their gameplay and moves in order to get better, paired with reflection cycles of trial-and-error (as done for their own replays). Both WATCHING GAMEPLAY and COPYING OTHERS were rated of high relevance for the three game genres and can be considered as other universal learning strategies.

In addition to videos as a medium of information, some interviewees revealed that they spent time finding specific BUILDS AND TACTICS online and following them to develop their skills. This included the choice of buildings and items, sequencing, leveling/upgrading, and more. Learning a good build order¹ was seen as a foundational element of learning the games at beginner levels. However, BUILDS AND TACTICS showed to be more genre dependent, rated significantly more important for learning to play RTS games (high relevance) than for MOBA and BR games (moderate relevance). They were considered the least (albeit still moderately) relevant for BR games which also exhibited significant differences to MOBA games.

Other means of analyzing and understanding games outside of a game itself were classified as EXTERNAL DATA TOOLS. While builds and

tactics provide general information on strategies, data tools use actual data from games to allow for a review of player performance. This kind of analysis was performed on community platforms as, for instance, OpenDota [59] or Sc2ReplayStats [60] which offer additional features such as the ability to generate heat maps of locations and movement. Such analysis was seen to be important for learning and showed high relevance for all the three genres with no significant differences.

Overall, there was a deep sense of the need of being dedicated, as observed from the interviews. Some of the interviewees emphasized the importance of creating a ROUTINE AND HABITS in the learning process. This included incorporating both mental and physical aspects in their routine to optimize efficiency, reduce the load on their working memory, and develop the right attitude to strive for from a learning perspective. The following comment elaborates on the importance of building routine:

StarCraft is a game of habits: You need to have good routines to get good at the game. This means simply a lot of playing, watching replays, analyze, and repeat. Daily, if you want to get really good. [S3]

These impressions were also confirmed by the survey, where routine was seen as highly relevant for BR and RTS games and moderately relevant for MOBA games but without significant differences (cf. Table 1).

Interestingly, ordinal regressions did not show any significant associations ($p \geq .05$) between the years playing and the ratings of the different categories, except in two cases: (1) Years of playing showed a significant relation with COPYING OTHERS in case of BR games, with the importance declining with increasing years of playing ($b = -0.64$, $SE = 0.28$, $Wald = 5.05$, $p = .025$). (2) Years of playing also had a negative relation with BUILDS AND TACTICS in case of MOBA games ($b = -0.16$, $SE = 0.07$, $Wald = 4.58$, $p = .032$).

4.2. Which aspects can impact self-efficacy (RQ 2)

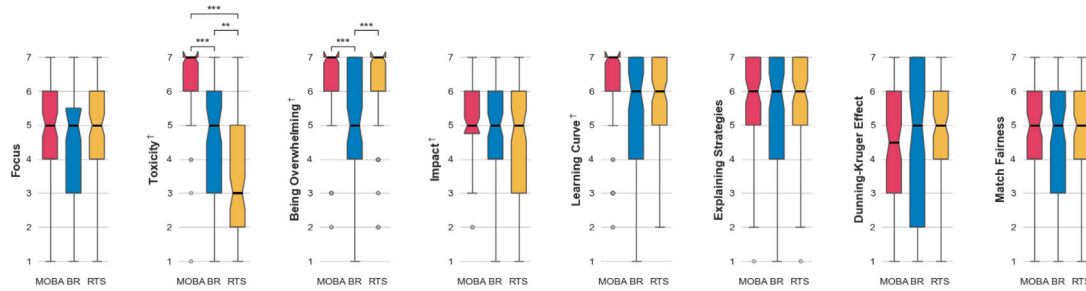
The results showed that self-efficacy was an important factor to influence players' motivation, especially in the beginning phase. A number of interviewees acknowledged that there was a steep LEARNING CURVE or a general feeling of these games BEING OVERWHELMING (at the beginning) due to the sheer amount of information which needed to be processed during gameplay. While the two are partly related (e.g., the learning curve may be steep due to too much information to be consumed), BEING OVERWHELMING is about the amount of information that needs to be processed at a moment in time in general, while LEARNING CURVE is rather focused on the overall difficulty (e.g., skill needed to succeed, others being better) to get started with a game. As such LEARNING CURVE, as opposed to BEING OVERWHELMING, is also more longitudinal in nature. Both often caused players to feel lost, in turn, affecting their self-efficacy and motivation. The following comment is a recollection of such an experience:

However, when my brother tried StarCraft, he didn't understand five things, which is four too many at a time. Not that he didn't want to but you need perseverance and patience to tackle these at a time, and you don't see progress as clearly and quickly. [S2]

These aspects were also endorsed in the survey, showing that a steep LEARNING CURVE can scare beginners² away and thus to be highly relevant for self-efficacy across the three genres. Differences between the genres were not significant (cf. Fig. 3). However, the feeling of BEING OVERWHELMING was found to be highly relevant for MOBA and RTS games, significantly more so than for BR games where moderate relevance was observed.

¹ A build order describes in which order different buildings should be constructed to establish a strong economy early on.

² A beginner player, as referred to by the interviewees, is someone who has recently started playing a game.



† Kruskal-Wallis H test significant at false discovery rate adjusted $p < .05$, Dunn's post-hoc test for multiple comparisons significant at $\alpha = .016$ (Bonferroni corrected), $\alpha = .01$, $\alpha = .001$

Fig. 3. Agreement on how relevant the developed categories on self-efficacy are across the different genres (1 = totally disagree, 7 = totally agree).

On the other hand, sometimes beginner players' behavior was described as being indicative of the DUNNING-KRUGER EFFECT [61]. The phenomenon describes that the ability of people to judge their competence depends on possessing the competence in question. As such, people with low competence at a task tend to overestimate their performance. In our context, interviewees described how beginner players often think they are much better than they are, caused by ignorance of the intricacies and nuances of the game. Even though this seems to contradict how other players feel overwhelmed initially, both aspects can eventually have a negative effect on self-efficacy of the players, caused by an inability to process vast amounts of game-related information and knowledge during the initial phases. This concern is shared in the following comment by D5, who has experienced the game both as a player and a coach and explicitly mentioned the effect by name:

I see the Dunning-Kruger effect in action with my students all the time. They spent 100 h in the game, and they think they are of the highest level. This can be good for confidence, but you have to watch out for the incredible disappointment that may follow when you find out there's still so much more. [D5]

However, there were fewer mentions related to the DUNNING-KRUGER EFFECT in the interviews than for the feeling of BEING OVERWHELMED for beginner players, which is also reflected in the survey results, showing a moderate relevance – without significant differences – across the genres.

In addition, beginner players' self efficacy was also affected by the TOXICITY within the gaming communities. In fact, toxicity was being identified as the second most important aspect in the interviews. This would manifest during the gameplay through hostility in the chat and other forms of communications with teammates, mainly caused by the competitiveness of the teammates but also by a misalignment of skills and expectations when beginner players were matched with advanced players having a lot more experience. It is worth noting that the toxic behavior mentioned in the interviews may not be reflective of the overall gaming communities but the personal experiences of the interviewees. The survey results showed considerable variation, with TOXICITY having high relevance for MOBAs, moderate relevance for BR games, and low relevance for RTS games, with significant differences between each.

While discussing about the issues related to self-efficacy, several strategies and mechanisms which can have a positive impact on the self-efficacy of players also emerged. Related to the complexity of the game and overwhelming information, players said that it was important to FOCUS on single aspects of the game and track their progress in it, giving them a sense of improvement. As the most frequently cited category, interviewees often justified how FOCUS helped them to actively internalize a prospected skill as well as to generate positive feedback that makes them feel good about their abilities. This is well illustrated by the following comment:

For example, we're talking about builds; and the goal is to not be supply blocked³ the first five minutes; that is very important for a good start of the game. So you would have this goal: Don't be supply blocked for the first x minutes. Then you see a timeline, when you were supply blocked. Then you can see how you did regarding your specific, concrete, goals. Then even if you lose, you can say: Hey, I did it! I achieved my goal to some success. It's also important to have a reference: your average [performance]. [S2]

Trying to FOCUS on single aspects of a game as a strategy to improve self-efficacy was, however, only found moderately relevant for all the three game genres in the survey, with no pronounced statistical differences.

Another suggestion that surfaced to improve self-efficacy was to EXPLAIN STRATEGIES for a game better and earlier in the game. Often not included in game tutorials, these strategies can include macro concepts beyond mechanics and control that would lead to insights and directions for making gameplay choices. For instance, how to create pressure on an enemy, how to time builds and upgrades, how to position armies, and similar. An improved sense and motivation for players was attributed to having a better grasp of the 'bigger picture'. Survey results also endorsed the importance of EXPLAINING STRATEGIES for self-efficacy, showing a high relevance across the three genres.

Other strategies linked to improving self-efficacy involved players not only looking at their own information but also at the data of other players with whom they play with. Almost half (=6) of the interviewees emphasized that it is important to understand the IMPACT of their actions and decisions in relation to their teammates, i.e. on the overall performance of the team, to increase their self-efficacy. The following comment highlights the value of showing IMPACT, especially for beginner players:

It's important to give people the feeling they have impact, that their efforts matter, even if they're new, or playing with better people, that they can still make game-deciding plays. [D3]

Survey findings indicate moderate relevance of showing one's IMPACT to improve self-efficacy for all the three game genres. While IMPACT is about being able to assess contributions within a team, some interviews showed that looking at the opponents' information was also important to perceive MATCH FAIRNESS. If the players know that they are matched against better players, a loss could be perceived as a learning opportunity and does not demotivate the players, thus affecting self-efficacy. Across all three genres, MATCH FAIRNESS showed to be of moderate importance.

Ordinal regressions did not show any significant association between the years playing and any of the self-efficacy categories.

³ A term usually used in RTS games to refer to instances when missing supplies halt unit production, likely delaying build order.

5. Discussion

Responding to the need of more player-centric research [2], the present study shares a detailed account of the various mechanisms and processes through which players learn in games as well as associated contextual factors that affect their learning (cf. Table 1) and self-efficacy (cf. Table 2). In addition, the results were examined for their relevance across the three studied genres (cf. Figs. 1 and 3). In the following, the salient aspects of learning and self-efficacy of players in these genres are discussed.

5.1. Learning

In this section, we discuss five broad characteristics of learning in the competitive games we investigated. These can serve as starting points for future research and design interventions aimed at supporting skill acquisition of players.

Learning happens largely by evaluating processes. Generally, our results suggest that players heavily rely on augmented feedback (i.e. feedback from an external source) for skill learning. Augmented feedback has been documented to have the ability to not only facilitate skill acquisition but also motivation [63]. Usually, augmented feedback is classified into *knowledge of results* (information about the performance outcome) and *knowledge of performance* (feedback about how an action was performed) [64]. The former is, for instance, evident in players' use of statistics websites to view and compare performance metrics (cf. EXTERNAL DATA TOOLS). The latter, however, seems to play an even greater role, with players paying attention to understanding the processes behind their performance. Like video recordings of one's own performance are an important source of knowledge of performance in sports (see [64]), WATCHING GAMEPLAY videos and reviewing REPLAYS take a central place across all three investigated genres. EXTERNAL DATA TOOLS can also be counted towards this category if they go beyond merely showing performance summaries or metrics. Likewise, as coaches have been identified as an important source of information in sports [64], players of video games learn from other players as reflected in categories such as COPYING OTHERS (more on this below).

However, researchers also report (cf. [65]) that too much feedback may lead to an overreliance and dependency on feedback on behalf of the learner. Learning support tools for players should thus deliver the right information at the right time, especially when feedback is provided in real-time. Future studies may thus probe deeper into which and how much information is valuable at which points in time (e.g., post-play, real-time) for skill development in, but not limited to, competitive games.

Learning strategies are largely independent of genres. Even though the investigated game genres were quite different in their gameplay, it was interesting to see that players identified and described similar mechanisms and processes for learning to play these games in their interviews. This was further corroborated through survey results where the majority of the categories related to learning (8 out of 12, see Fig. 1) were rated to be of high relevance with no statistical differences between the three genres. This pattern could be reflective of certain meta-level processes, presenting opportunities for research, the design of tutorials, ways of teaching throughout play, and training visualizations. Hesketh et al. [1] also reported that players attached importance to meta-cognitive and meta-learning processes and skills, for example, identifying knowledge gaps, deliberation, or learning to learn.

However, there were also some exceptions which showed statistically significant differences in relevance between genres, specifically BUILDS AND TACTICS, PHYSICAL DEXTERITY, CAMPAIGN, and AI. This can likely partly be attributed to differences in the game mechanics (e.g., speed of gameplay, controlling a single character vs multiple units, long-term planning vs. short-term agility) of the investigated games and their strong multiplayer focus.

BUILDS AND TACTICS (construction-related strategy) were highly relevant for RTS games, followed by MOBA games, and lastly BR games. In RTS games such as *Starcraft II*, optimizing build orders is important to establish a healthy economy early on the game to be able to build a strong army in order to be able to succeed. Build orders are thus heavily discussed on community websites such as *Liquipedia* [66]. Similarly, champion builds are important in MOBAs to maximize their effectivity. BR games, on the other hand, do not put that much emphasis on builds. Improving PHYSICAL DEXTERITY (aiming and movement related skills) had significantly higher relevance in BR games compared to the other two investigated genres. This can likely be explained by their shooter-like gameplay and consequently the importance to be able to adequately control the avatars actions (cf. [67]) to react quickly to threats. Such games hence require good hand-eye coordination and thus sensorimotor skills. Indeed, Park et al. [68] found that professional *Counter-Strike: Global Offensive* players possess higher elbow and wrist rotation than amateur players. While the other two genres also require proper sensorimotor skills, their gameplay might make demands comparable smaller.

The other two categories of CAMPAIGN and AI represent specific game modes/features of games which in our case are most pronounced in RTS games. The RTS games to which the survey answers related to (*StarCraft*, *Age of Empires*, and *Command & Conquer*) all feature single-player campaigns that also act as a way to gradually introduce gameplay mechanics and units. MOBA and BR games, on the other hand, usually are focused on multiplayer with only marginal single player modes. Likewise, playing against AI opponents is considered more useful in RTS games, followed by MOBA and lastly BR games. While the latter two also offer bot modes, playing against an AI appears much more common in the case of RTS. These differences could perhaps be attributed to the different quality of AI bots across games. This would also be an explanation why the value of playing against an AI for learning is generally rated rather low: AI bots might not operate on the skill level necessary to compete against real players.

In summary, learning seems largely independent of the genre but genre-specific mechanics put emphasis on different aspects, at least in the ones investigated here.

Learning is social. Social learning theory posits that learning new behaviors largely occurs by imitating and observing others [69]. This is also reflected in our results. The interview and survey results highlighted a substantial social element of learning to play, including aspects such as playing with FRIENDS, WATCHING GAMEPLAY, COPYING OTHERS, and trying BUILDS AND TACTICS from others. This need for learning socially has led to strong community-driven initiatives that offer access to play-related data (e.g., BUILDS AND TACTICS) through EXTERNAL DATA TOOLS on platforms outside of the game. These initiatives have also been recognized as out-of-game learning tools in recent research on learning in esports [1] and include streaming services (e.g., *Twitch.tv*, *YouTube*), forums (e.g., *Reddit*, *Steam*), out-of-game communication tools (e.g., *Discord*), and statistics services (e.g., *OpenDota* [59]). Kleinman and El-Nasr [2] also acknowledged the value of data-driven services for training in esports.

This also suggests that players form *communities of practise*, i.e. *groups of people who share a concern or a passion for something they do and learn how to do it better as they interact regularly* [70], in order to learn from each other. Such communities of practise have three defining characteristics (cf. [70]) which we see reflected in our results: players (1) operate in a shared domain of interest, (2) engage in both sustained joint activities and discussions, and (3) in the creation of shared resources (e.g., through the sharing of builds and tactics or through creating and using data platforms and websites).

One could also say that learning to play extends beyond the actual act of playing. Furthermore, even with seemingly self-focused learning activities such as reviewing their own REPLAYS to learn from their mistakes, players preferred to invite their friends and teammates to reflect

together, as seen in the interviews. Individual learning, in that sense, is closely related to learning together, raising an interesting direction for future research, focusing on how learning and training visualizations can cater to teams as a collective, and not just individual players. High-level planning accounting for team actions as well as trying to improve and evolve by adopting the ‘best’ strategies by taking inspiration from the global community has been identified as a key learning goal for esports players [1]. It is worth noting that this discussion is situated and specific to games with a popular multiplayer game mode, as was the case for the games included in this study. Nevertheless, the social learning aspect of games has also been acknowledged in single-player and co-located multiplayer games through strategies such as knowledge exchange and guidance between the players [21].

Learning is highly reflective. There was an overall display of a critical and reflective attitude by the players when analyzing their REPLAYS or when WATCHING GAMEPLAY of other players to learn from it. Interviewees also made reoccurring references to how they consciously alter variables of the game to understand how it affects their performance, a strategy resembling Korthagen’s ALACT Model of Reflection [62]. This aspect was captured under the category of CONSCIOUS REFLECTION, rated highest by players in terms of its relevance for learning all three examined genres. Iacovides et al. [21] in their research on players’ learning strategies when getting *breakthrough* insights in games made a distinction between “trial-and-error” and “experiment” strategies, with the latter being methodical and consciously testing a hypothesis through actions, leading to intended learning and not just progress in the game. Their experiment strategy corresponds well to our CONSCIOUS REFLECTION category, emphasizing the reflective and intentional nature of the learning activity. Hesketh et al. [1] identifies the reflective learning process of the players as the meta-process of “deliberation”, stating that what players learned through different learning mechanisms and processes was dependent on this meta-process. Kleinman et al. [71] observed that advanced players also showed more structured practise routines than novice players.

So-called *deliberate practise* [72], a structured and purposeful activity that is specifically directed towards improving performance, has been reported to play an important role in the development of expertise in various domains (cf. [73]). Our findings and similar observations made in the above mentioned related studies provide evidence that players also engage in deliberate practise to reach higher levels of in-game performance. Consequently, our results show that there are opportunities for game developers and researchers by focusing on identifying ways that support reflection and practice processes for players over time, in-game, or through additional support tools. However, if implemented as part of a game itself, this needs to be carefully considered as Towne et al. [74] discuss how the principles of deliberate practise may be conflicting with those of game design, as deliberate practise must not necessarily need to be enjoyable.

Learning to play requires appropriate teaching facilities. Learning to play is affected by how games teach to play. Long [75] outlined three approaches contemporary games use to teach players: through explanation, by letting them explore, and by relying on existing knowledge. Being too prescriptive curtails the ways in which players can learn a game. A game which, for instance, only relies on teaching through telling limits the possibilities to learn through reflective trial-and-error. However, learning to play is also what makes games ‘fun’ to play [75]. In this sense, our results can be informative for the design of in-game teaching systems (e.g., tutorials).

These possibilities extend beyond the game itself by providing off-game platforms that facilitate learning and, in turn, keep players engaged with games. The strong emphasis on out-of-game learning strategies could have its origin in the competitive nature of the studied genres which require dedicated effort to learn and improve, unlike single-player and co-located games where learning strategies seem to be linked with the actual playing time, at least in the study of Iacovides

et al. [21]. For example, both WATCHING GAMEPLAY and the use of EXTERNAL DATA TOOLS were considered highly relevant in our study. Both of these learning strategies have been mentioned as theoretical propositions in existing literature [3,37,40,41] which our results empirically corroborate.

Knowledge exchange using external modalities such as sharing strategies has also shown to be important for learning as well as self-efficacy (cf. BUILDS AND TACTICS and EXPLAINING STRATEGIES). These processes can be facilitated by game developers by, for example, offering public APIs to give access to in-game data for creating statistics websites and similar. Offering appropriate means to foster learning of games outside of them can also contribute to self-efficacy and, in turn, retention. For instance, the impact a player makes cannot only be communicated through in-game feedback but also through post-play visualizations.

5.2. Self-efficacy

As mentioned in Section 2, Bandura [11] identified four informal sources of self-efficacy. Two of them – mastery experiences (gained through performing a task) and social persuasion (verbal encouragement) – are reflected in our categories. BEING OVERWHELMING, LEARNING CURVE, MATCH FAIRNESS, IMPACT, as well as being able to FOCUS on single aspects all are related to factors that influence play and thus impact one’s ability to gain mastery experiences through practice. TOXICITY can be connected with negatively influencing social persuasion. Vicarious experiences (observing of others), as third source of self-efficacy, was not explicitly mentioned when asked about self-efficacy but formed an important opportunity for learning to play (cf. WATCHING GAMEPLAY). Interviewees may not have been consciously aware how learning through watching can help build self-efficacy. Interestingly, none of the categories was related to the fourth source: the physiological and emotional state of a person.

Generally, and similar to the categories about learning, the identified factors influencing self-efficacy were rated of similar importance across the genres (apart from TOXICITY and BEING OVERWHELMING, cf. Table 2), suggesting that factors affecting self-efficacy are, to a large extent, independent of genre. However, genre-specific factors can promote or diminish self-efficacy. For example, the differences in BEING OVERWHELMING can likely be attributed to the varying complexity of the included game genres in terms of how much information and how many tasks players have to concurrently process and take care of. Both MOBAs and RTSs games were rated significantly higher in terms of being overwhelming. In both games, players have to monitor the whole map to react quickly and take appropriate countermeasures. RTS games also require players to manage and coordinate up to hundreds of units, while MOBAs necessitate strong team coordination and knowledge of different champions and abilities. While also supporting team modes, the gameplay of BR-like games is more focused on the immediate surroundings. In connection to this, the viewpoint could also be hypothesized to have an impact. Both RTS and MOBA games use an isometric viewpoint to show a larger fraction of the environment, while BR games offer first- and third-person cameras which show the current field of view of the character only.

In the following, we discuss three characteristics of self-efficacy that manifested in our analysis.

Self-efficacy is especially important in the early phases of play. Our findings support the role of self-efficacy in affecting players’ motivation in video games [12]. Acknowledging the role of self-efficacy in affecting motivation and learning [11,42], interviews showed that self-efficacy was critical, especially for beginner players in the early phases of play. This association was attributed to the high difficulty players had to face in the initial playing phase caused by a lack of understanding of the nuances of a game. This would manifest in different ways such as the playing experience BEING OVERWHELMING for some players or in a steep LEARNING CURVE, while others overestimate their abilities (DUNNING-KRUGER

EFFECT, cf. [61]). The survey results showed that the LEARNING CURVE was highly relevant while the seemingly contrasting DUNNING-KRUGER EFFECT was only moderately relevant for players from all the game genres. In both cases, players would tend to be more vulnerable and likely to lose motivation at some point of their initial journey, unless they could track their real progress leading to higher self-efficacy. Beginner players could thus benefit from data trackers which explicitly consider ways to enhance self-efficacy. Likewise, in-game training systems (e.g., tutorials) that focus on new players could benefit from facilitating mechanisms that contribute to perceived self-efficacy.

Self-efficacy is influenced by the community. Self-efficacy is not only a matter of one's own skills and performance but is also influenced by others. In our study, toxicity has emerged as a major factor in undermining self-efficacy. However, the impact of toxicity varied considerably between genres, with MOBAs scoring statistically significantly higher than BRs which, in turn, scored significantly higher than RTS games. This could be a consequence of the varying toxicity in these kind of games. Indeed, a study by Tyack et al. [76] has found that unsportsmanlike behavior was the reason for about every tenth player to stop playing MOBAs. However, issues of toxicity are not unique to MOBAs but also appear in BR (e.g., [77]) and RTS games (e.g., [78]). While the reasons for the different ratings could be manifold, we suspect that the different levels of dependence on others in the three genres could be one explanation. MOBA games are almost exclusively multiplayer games where team work and coordination are crucial, BR-like games also allow players to play alone against others (in addition to teams), and in RTS games (while also offering modes such as 2vs2) players usually compete one on one in the competitive scene. Different communication tools offered by the games (e.g., text chat, voice chat, pings) could be another explanatory variable.

While the influence of deviant behavior on retention is manifold, we would argue that its diminishing effect on self-efficacy could be one reason for it. Bandura [11] has identified verbal persuasion (i.e. encouragements regarding one's performance) as one of four informational sources for forming self-efficacy beliefs as mentioned previously. In this sense, toxic behavior can be discouraging and hence detrimental to building self-efficacy.

Self-efficacy can be supported. Lastly, our results indicate that the self-efficacy of players can be raised through providing feedback that is also explanatory. Participants discussed how getting a feel of the IMPACT one has and how EXPLAINING STRATEGIES, i.e. macro concepts, support self-efficacy. Offering feedback in the form of the impact a player had, in that sense, could thus further contribute to the value of enactive mastery experiences. Studies investigating means to foster self-efficacy through feedback are, however, currently scarce (a notable exception being Jamshidifarsani et al. [79]). As such we would advocate for more research in this space as there may be many untapped opportunities to specifically design for fostering self-efficacy in games. Providing evidence of the fairness of matchmaking can be helpful as well. Self-efficacy is also shaped by the self-perceived difficulty of a task (cf. [80]) – being matched, for instance, with comparatively too inexperienced players could thus weaken self-efficacy.

5.3. Implications

In this work we have identified different resources, tools, and strategies players use to learn to play in and outside of a game and how self-efficacy beliefs of players are influenced.

Notable, years of playing a game did in general, except for two instances, not show to have had a significant impact on how important the different categories for learning and self-efficacy were considered. Yet, the two which were significant are reasonable: the value of COPYING OTHERS to decline in case of BR games appears sensible as once a player improves and becomes better themselves copying from others may become less important and rather others may copy from them. The

decline of importance of BUILDS AND TACTICS in case of MOBAs could be explained by their importance to be successful and thus players learning them earlier on and memorizing them. Anyway, it would be expected that the value of more categories changes across genres with growing experience. We highly assume, that years of playing is not a good proxy for the experience a player has, as skill development is individual and some players may require more or less time to acquire the same skill level. Performance metrics such as ranks or league points in case of *League of Legends* or the matchmaking ranking of *Starcraft II* would be more appropriate measures but these are not comparable across games and genres and thus were not suitable for our genre comparison. That said, this aspect certainly warrants further dedicated investigations in the future.

Based on the identified categories we discussed different characteristics of how learning and self-efficacy manifests in three popular competitive genres. As such we do not claim completeness and care should be taken to not over-generalize our findings but rather view them within the sphere of the three investigated genres. Nevertheless, our work provides a further incremental step towards developing a more complete understanding about the processes players use to learn to play, reaffirming and expanding findings of initial works such as [1, 21] in this space. As such we extend the discussion on how player learn to play, which also became more pressing with the need to understand how players gain experience in esports (cf. [2]). In this sense, we believe our results can provide useful pointers on which aspects to study more closely in the future such as identifying ways on how reflective practise processes can be deliberately promoted through design. We could also see that self-efficacy beliefs are shaped by a variety of factors, pointing to a complex phenomenon, that would deserve further consideration of self-efficacy theories in game design.

Beyond researchers, our work can help game developers to create game experiences that support learning and self-efficacy. Both have been argued to be important for the continued engagement with a game. Our work provides an indication on which concrete factors to focus on to support both. For example, through offering resources and services (e.g., analysis tools, guides) that match learning preferences. Our results also point to an apparently strong social aspect to skill development which could be facilitated by providing respective features directly within a game. For instance, in the game *Destiny*, players volunteer as so-called *sherpas* that assist others with completing raids [81]. Efforts could be made to support such community efforts directly within a game.

Moreover, our results could bear value for community managers who can create means that specifically support learning endeavours (e.g., through highlighting videos that demonstrate certain skills well and can serve as best-practice examples) and counteract community behaviors detrimental to self-efficacy (e.g., toxicity).

5.4. Limitations and future work

Our study was conducted with a focus on competitive multiplayer games. While we anticipate a certain generalizability across games – also due to partly overlapping identified learning strategies in previous work [1,21] as discussed above – we would still advise caution in extrapolating these findings to other genres, especially single-player games. For instance, playing against the AI or through the CAMPAIGN have shown to be of low relevance for learning in the three genres under investigation. Both are not common in BR and MOBA games. For RTS games, which usually feature a campaign and opportunities to play matches against an AI, ratings were – albeit still low – higher. As such it might be reasonable to assume that both play a larger role in games where the two features are more pronounced. We also did not probe into *why* certain differences in the relevance for some categories did unearth. However, we provided hypothesis on potential factors to guide future studies. Such studies should investigate the reasons why

some of the aspects are considered more or less relevant for specific genres through qualitative inquiry.

As the interviews were based on unbalanced interview groups, with BR being represented by about half as many interviewees as the other two genres, the developed categories may be skewed in favor of MOBA and RTS games. As such, categories of high-relevance in BR games but not in the other two genres may not have been captured to the same extent as vice-versa. In other words, certain nuances and specific factors unique to BR games might not be fully represented in our analysis. As such the categories described here should not be seen as exhaustive. That said, the relevance ratings obtained via the survey are based on almost even sample sizes (BR: $n = 83$, MOBA: $n = 80$; RTS: $n = 101$).

We also only looked at genre-wise comparisons and did not explore learning specific to individual game mechanics or actions within the studied genres. Future work could explore differences based on specific mechanics within genres but we would also like to reiterate that the identified categories have shown to be mostly independent of the genres themselves.

Kleinman et al. [71], studying how self-regulated learning manifests in the context of esports, noted that players seem to shift from process to outcome goals and become more structured as they become more skilled. Advanced players also showed more structured practise routines than novice players. While interviews were based on players with diverse backgrounds and levels of expertise to capture potentially different mechanisms, the survey responses did not show – except for two cases – differences in the relevance of the identified ways for acquiring game-related skills based on expertise level (that is, years playing the game). Investigating this further, would be an interesting avenue for future research. As mentioned earlier, self-regulated learning and self-efficacy form a reciprocal relationship. In this study, we derived categories with respect to both but we did not examine how they are related to each other. In the future it would thus be of interest to more closely observe how these are interwoven.

Our sample was heavily skewed towards males. This is to some extent reflective of the small percentage of female gamers in the genres considered here (e.g., a survey [82] among 270,000 players found that only 10% of players of MOBAs are females, even less so for FPS). As such and since work on learning in games in general and specifically with respect to gender is scarce, more studies would be needed to explore gender differences. That said, previous research, as discussed by Severiens and Dam [83], has also shown that gender differences in learning styles are small on average, but results vary considerably across studies. Blumberg and Sokol [35], for instance, found no differences in girls and boys relying on internal or external strategies to learn to play video games, compared to age. However, in respect with self-efficacy, gender differences appear more pronounced with males showing in general a greater sense of self-efficacy (cf. [7]). Research has also shown that men tend to have higher self-efficacy in relation to video games [84]. It is thus possible, that a more diverse sample might report additional factors that inhibit or enhance self-efficacy. Due to the gender imbalance, we also did explore potential differences in skill levels across genders. Besides gender, future studies may also want to take other variables such as age, expertise, or the role a player is taking on in a team into account.

Lastly, while we took measures to control for the large numbers of statistical tests (controlling for false-discovery rate and adjusting p -values for multiple comparisons), differences between genres should be interpreted as indicative rather than confirmatory.

6. Conclusions

In order to make informed choices and act and react meaningfully, players need to develop an understanding of the intricacies of a game. Game developers can facilitate this process by implementing appropriate teaching mechanisms, within (e.g., tutorials) and outside a game (e.g., strategy guides). This, however, requires an understanding

of how players gather knowledge and develop their skills. Our study contributes to this currently sparsely explored but important aspect of player interaction by offering an overview of (a) the different ways how players learn to play and (b) factors contributing positively or negatively to self-efficacy.

Our results indicate that both learning and self-efficacy is largely independent across the competitive multiplayer genres (MOBA, BR, RTS) we investigated. In addition, learning has shown to be a reflective and social process extending beyond the act of play itself. Perceived self-efficacy can especially be affected in the early phases of play and influenced by a game's community but also be promoted by providing support, for instance, in the form of advice.

Future work may probe deeper into the individual factors identified here to enrich our understanding of the processes and resources involved when learning to play.

CRedit authorship contribution statement

Günter Wallner: Conceptualization, Formal analysis, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Aakash Johry:** Formal analysis, Writing – original draft, Writing – review & editing. **Marnix van Wijland:** Conceptualization, Data curation, Formal analysis. **Regina Bernhaupt:** Conceptualization, Methodology, Writing – review & editing. **Simone Kriglstein:** Conceptualization, Formal analysis, Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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